Diagnosis

Diagnosis of PKU is made by documenting elevated levels of phenylalanine in the serum (20 mg/dL or higher, 10 to 50 times the normal level). Plasma tyrosine levels may be normal or elevated. In addition, elevated urinary phenylpyruvic acid levels can be detected; this is often confirmed by a characteristic green coloration of urine after the addition of 10 percent ferric chloride. Other laboratory findings include phenylalanine hydroxylase deficiency, phenylpyruvic acidemia, and increased urinary excretion of o-hydroxyphenylacetic acid and phenylacetylglutamine.

Newborn screening programs in many countries use either the semi-quantitative bacterial inhibition assay (Guthrie test), which detects elevated phenylalanine levels from a few drops of capillary blood, or a quantitative chemical method (Quantase test). In affected infants, serum phenylalanine levels typically begin to rise by the third or fourth day of life. Prenatal diagnosis is possible via amniocentesis or chorionic villus sampling with genetic analysis. Preimplantation genetic diagnosis also allows for the selection of embryos unaffected by PKU, enabling the birth of PKU-free children.

Pathophysiology and Histology

Classic PKU results from a deficiency of phenylalanine hydroxylase or its cofactor, tetrahydrobiopterin, leading to accumulation of phenylalanine. The normal phenylalanine-to-tyrosine ratio is approximately 1:1, but in PKU it exceeds 3:1. The observed pigment dilution is thought to result from phenylalanine's inhibitory effect on tyrosinase, an enzyme critical for melanin synthesis.

Skin biopsy reveals increased fibroblasts and histiocytes, atrophy of skin appendages, and sparse, fragmented elastic fibers—distinct from the changes seen in true scleroderma. Histologically, melanocyte numbers are normal, but there is a reduction in mature melanosomes.

Differential Diagnosis

PKU must be differentiated from primary atopic dermatitis and scleroderma (Box 131-3). Other conditions to consider include disorders of amino acid metabolism with neurocutaneous manifestations. Transient elevations in plasma phenylalanine may also occur in premature infants and individuals with liver disease.

Box 131-3

Differential Diagnosis of Phenylketonuria

Main Features

- **Skin**: Pale pigmentation, dry skin, eczema, scleroderma-like lesions, fair hair
- **Eyes**: Blue eyes
- **Neurologic features**: Mental retardation, seizures, irritability, increased deep tendon reflexes

Consider

- Other disorders of amino acid metabolism with neurocutaneous features

Always Rule Out

- Atopic dermatitis
- Scleroderma

Treatment

A low-phenylalanine diet, ideally maintained for life, leads to reversal of skin findings such as hypopigmentation, photosensitivity, musty odor, and eczema. Early initiation of dietary therapy significantly reduces the risk of mental retardation, and maintaining blood phenylalanine within acceptable limits during early childhood supports normal to low-normal cognitive development (Box 131-4).

However, strict dietary control may lead to protein deficiency and associated eczematous dermatitis.

Box 131-4

Treatment of Phenylketonuria

- Low-phenylalanine diet
- Fish oil supplementation (may improve visual evoked potentials)
- Tyrosine supplementation (?)
- New phenylalanine-free formulas
- Reduction of phenylalanine uptake using purified phenylalanine ammonia-lyase
- Recombinant or pegylated phenylalanine ammonia-lyase
- Tetrahydrobiopterin supplementation (to enhance residual phenylalanine hydroxylase activity)
- Supplementation with large neutral amino acids (to compete for transport across the blood-brain barrier)